

Caroline DARCHE-OLSSON

Engineer in Molecular and Cellular Biology

Career Objective

Join an R&D team in biotechnology, pharma or cosmetics and contribute to the development of versatile and innovative projects.

Contact

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LinkedIn

My portfolio

Languages

English (*Fluent - C1*)

French (*Native*)

Spanish (*Bilingual*)

Interests

Gym · Nature · Restaurants · Programming

Professional Experience

Feb. – Sept. 2026, ExAdEx Innov, Biotech CRO, Nice

Biology Engineer

- Managed 6 client studies on *in vitro* models, from experimental work to report writing and data analysis.
- Contributed to 3 R&D projects: model characterization, compound evaluation and development of new experimental approaches.
- Developed the Longevity Lab and ExAdEx inventory digital tools.
- Managed the laboratory, stock and orders.

May 2023 – Jul. 2025, Institut de Biologie Valrose, CNRS Nice

Research Engineer in Molecular and Cellular Biology

- Wrote and optimized protocols to study protein–ion channel interactions in colorectal cancer.
- Cell culture, molecular biology, biochemistry and microscopy.
- Analysis and interpretation of results.
- Laboratory and stock management, training of new staff.

Master's Internships

- M2, CNRS Lyon – LBTI. Study of mutations associated with sarcoidosis.
- M1, University of Seville. Protein aggregation in *C. elegans*.

Web Applications

Longevity Lab ↗

Tool to visualize the effect of active ingredients on tissue aging signatures

ExAdEx Inventory

Migration of an Excel inventory to a web platform: stock, orders, locations, interactive agent, history

Technical Expertise

Programming & Coding

- Design and development of custom web tools and applications (data visualization, internal platforms) tailored to the team's needs
- Tools: GitHub, VS Code, Android Studio
- Self-taught skills in HTML, CSS and JavaScript

Study & Laboratory Management

- Design, planning and management of client and R&D studies
- Writing and improving SOPs
- GMP/GLP practices and quality control
- Sample traceability
- Analysis and interpretation of results
- Training of new staff
- Stock and order management

Experimental Techniques

- 2D cell culture and 3D spheroids
- *Ex vivo* human adipose tissue models
- Cell transfection
- RNA extraction
- RT-qPCR
- ELISA and cell viability assays
- Cloning and site-directed mutagenesis
- SDS-PAGE, Western blot and co-IP
- GST pull-down
- Protein purification
- Flow cytometry
- Seahorse XF metabolic assays
- Immunofluorescence
- Confocal microscopy
- PLA – Proximity Ligation Assay
- Microbiology
- ImageJ, Prism, Microsoft Office

Academic Background

2022 — Master's in Molecular and Cellular Biology, Cell Genetics and Pathology track
Université Claude Bernard Lyon 1

2020 — International Bachelor's in Biology
Boston University (USA) and Université Grenoble Alpes